



Port City International University

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Report on

Topology; Interface terminal with CLI commands; OSPF Protocol; EIGRP protocol.

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Experiment No: 01

Experiment Name: Topology Analysis

Introduction: network topology created in Cisco Packet Tracer that combines bus and ring structures using hubs, switches, and PCs. The purpose is to understand how data moves across both shared and switched segments and how the mixed design affects performance and reliability. The analysis focuses on device roles, data flow, fault points, and the overall efficiency of this topology.

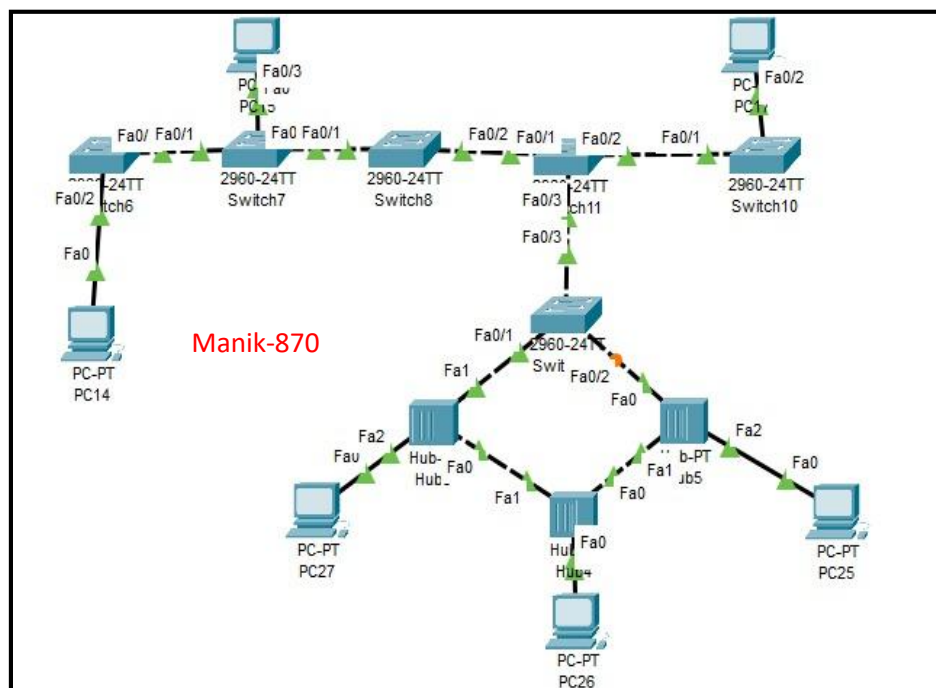
Software Requirement:

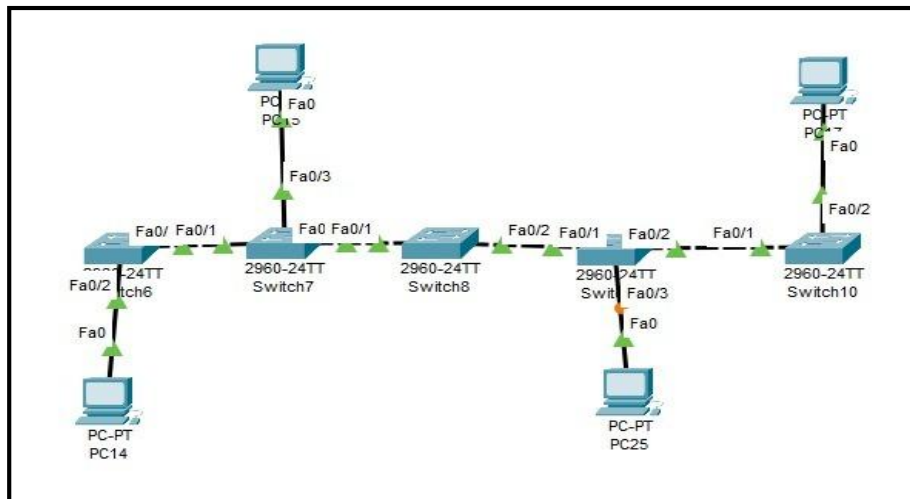
Cisco Packet Tracer – for designing, configuring, and testing the hybrid topology.

Procedure:

- Launch Cisco Packet Tracer and create a new project.
- Place required devices (PCs, hubs, switches) in the workspace.
- Construct the hybrid topology by connecting devices with appropriate cables.
- Configure IP addresses for all PCs.
- Verify link status and ensure all interfaces are active.
- Test basic connectivity using the ping command.
- Send a message using the Simple PDU tool to confirm end-to-end communication.

Output:





Discussion: The hybrid topology demonstrated how data flows across bus and ring segments. The bus segment shared a single medium, which can cause collisions, while the ring segment allowed more organized and reliable communication. Simple PDU (Protocol Data Unit) tests confirmed end-to-end connectivity. The experiment highlighted the importance of proper device placement, IP configuration, and link verification to ensure efficient network performance.

Experiment No: 02

Experiment Name: Interface terminal of a router in packet tracer with CLI commands.

Introduction: This experiment demonstrates the configuration of router interfaces using the Command Line Interface (CLI) in Cisco Packet Tracer. The goal is to assign IP addresses, enable interfaces, and verify connectivity between devices. Through this exercise, students learn the basic commands required to set up and manage router interfaces in a network.

Software Requirement:

Cisco Packet Tracer – for designing, configuring, and testing the hybrid topology.

Procedure:

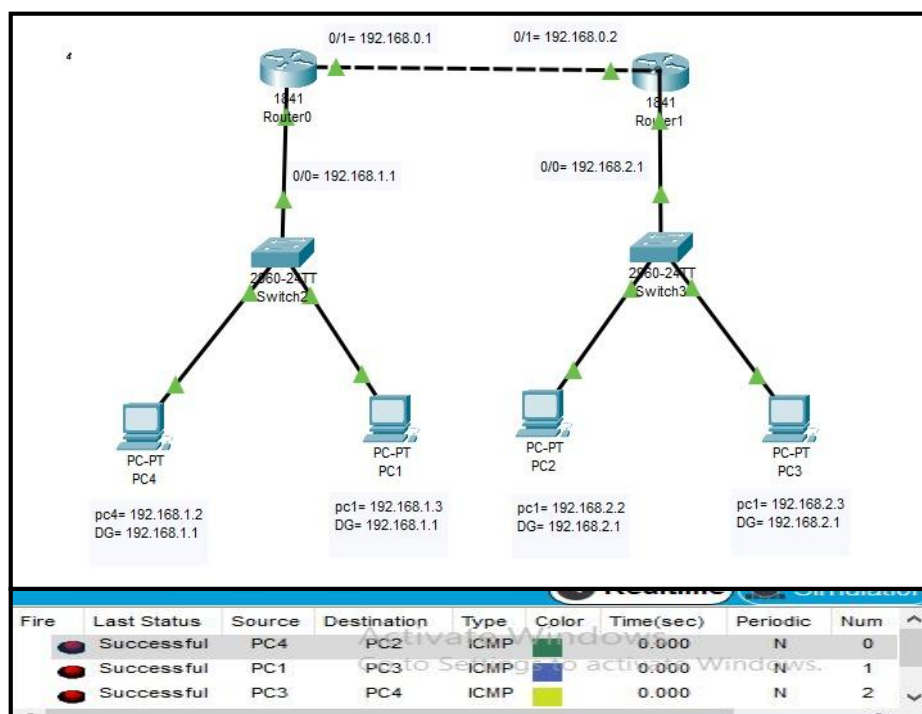
- Launch Cisco Packet Tracer and create a new project.
- Place routers and PCs in the workspace.
- Connect devices using appropriate cables.
- Access the router CLI:

Click the router → CLI tab.

Command prompt

Router0	Router1
<pre>Router>enable Router#configure terminal Router(config)#interface FastEthernet0/0 Router(config-if)#ip address 192.168.1.1 255.255.255.0 Router(config-if)#ip address 192.168.1.1 255.255.255.0 Router(config-if)#no shutdown Router(config-if)#exit Router(config)#interface FastEthernet1/0 Router(config-if)#ip address 192.168.0.1 255.255.255.0 Router(config-if)#no shutdown Router(config-if)#exit Router(config-router)#router rip Router(config-router)#network 192.168.1.0 Router(config-router)#network 192.168.0.0 Router(config-router)#exit</pre>	<pre>Router>enable Router#configure terminal Router(config)#interface FastEthernet0/0 Router(config-if)#ip address 192.168.2.1 255.255.255.0 Router(config-if)#ip address 192.168.2.1 255.255.255.0 Router(config-if)#no shutdown Router(config-if)#exit Router(config)#interface FastEthernet1/0 Router(config-if)#ip address 192.168.0.2 255.255.255.0 Router(config-if)#no shutdown Router(config-if)#exit Router(config)#router rip Router(config-router)#network 192.168.0.0 Router(config-router)#network 192.168.2.0 Router(config-router)#exit</pre>

- Test basic connectivity using the ping command.
- Send a message using the Simple PDU tool to confirm end-to-end communication.

Output:

Router0	Router1
<pre> Router>enable Router# Router#configure terminal Enter configuration commands, one per line. End with CNTL/Z. Router(config)#interface FastEthernet0/0 Router(config-if)#ip address 192.168.2.1 255.255.255.0 Router(config-if)#ip address 192.168.2.1 255.255.255.0 Router(config-if)#no shutdown Router(config-if)# %LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up Router(config-if)#exit Router(config)#interface FastEthernet1/0 Router(config-if)#ip address 192.168.0.2 255.255.255.0 Router(config-if)#no shutdown Router(config-if)# %LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up Router(config-if)#exit Router(config)#router rip Router(config-router)#network 192.168.0.0 Router(config-router)#network 192.168.2.0 Router(config-router)#exit </pre>	<pre> Router>enable Router# Router#configure terminal Enter configuration commands, one per line. End with CNTL/Z. Router(config)#interface FastEthernet0/0 Router(config-if)#ip address 192.168.2.1 255.255.255.0 Router(config-if)#ip address 192.168.2.1 255.255.255.0 Router(config-if)#no shutdown Router(config-if)# %LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up Router(config-if)#exit Router(config)#interface FastEthernet1/0 Router(config-if)#ip address 192.168.0.2 255.255.255.0 Router(config-if)#no shutdown Router(config-if)# %LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up Router(config-if)#exit Router(config)#router rip Router(config-router)#network 192.168.0.0 Router(config-router)#network 192.168.2.0 Router(config-router)#exit </pre>

Discussion: Configuring router interfaces via CLI allows precise control over IP addressing and interface status. Enabling interfaces and assigning correct IP addresses ensures that devices can

communicate across networks. Using CLI also introduces students to real-world router management practices, including interface verification and troubleshooting.

Experiment No: 03

Experiment Name: Implementation of a Network using OSPF Protocol.

Introduction: This experiment demonstrates the implementation of the Open Shortest Path First (OSPF) routing protocol in a network designed in Cisco Packet Tracer. OSPF is a dynamic routing protocol that enables routers to automatically exchange routing information and determine the shortest path for data delivery. The objective is to configure OSPF, verify routing tables, and ensure seamless communication between multiple network segments.

Software Requirement:

Cisco Packet Tracer – for designing, configuring, and testing the hybrid topology.

Procedure:

- Launch Cisco Packet Tracer and create a new project.
- Place routers, switches, and PCs in the workspace.
- Connect devices using appropriate cables.
- Assign IP addresses to all interfaces of the routers and PCs.
- Access router CLI and enter privileged mode:

Router0	Router1	Router2
Router>enable Router#configure terminal Router(config)#interface serial2/0 Router(config-if)#ip address 20.0.0.1 255.0.0.0 Router(config-if)#clock rate 64000 Router(config-if)#no shutdown Router(config-if)#exit Router(config)#interface serial3/0 Router(config-if)#ip address 10.0.0.1 255.0.0.0 Router(config-if)#clock rate 64000 Router(config-if)#no shutdown Router(config-if)#exit Router(config)#router ospf 1	Router>enable Router#configure terminal Router(config)#interface serial2/0 Router(config-if)#ip address 10.0.0.2 255.0.0.0 Router(config-if)#no clock rate Router(config-if)#no shutdown Router(config-if)#exit Router(config)#interface serial3/0 Router(config-if)#ip address 30.0.0.2 255.0.0.0 Router(config-if)#no clock rate Router(config-if)#no shutdown Router(config-if)#exit Router(config)#router ospf 1	Router>enable Router#configure terminal Router(config)#interface serial2/0 Router(config-if)#ip address 20.0.0.2 255.0.0.0 Router(config-if)#no clock rate Router(config-if)#no shutdown Router(config-if)#exit Router(config)#interface serial3/0 Router(config-if)#ip address 30.0.0.1 255.0.0.0 Router(config-if)#clock rate 64000 Router(config-if)#no shutdown Router(config-if)#exit Router(config)#router ospf 1

Router(config-router)#network 192.168.1.0 0.0.0.255 area 0 Router(config-router)#network 10.0.0.0 0.255.255.255 area 0 Router(config-router)#network 20.0.0.0 0.255.255.255 area 0 Router(config-router)#exit	Router(config-router)#network 192.168.2.0 0.0.0.255 area 0 Router(config-router)#network 10.0.0.0 0.255.255.255 area 0 Router(config-router)#network 30.0.0.0 0.255.255.255 area 0 Router(config-router)#exit Router(config)#	Router(config-router)#network 20.0.0.0 0.255.255.255 area 0 Router(config-router)#network 30.0.0.0 0.255.255.255 area 0 Router(config-router)#exit
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- Test basic connectivity using the ping command.
- Send a message using the Simple PDU tool to confirm end-to-end communication.

Output:

network= 10.0.0.0

Se3/0 = 10.0.0.1

Se2/0 = 10.0.0.2

Se3/0 = 30.0.0.1

Se2/0 = 30.0.0.2

Se3/0 = 20.0.0.1

Se2/0 = 20.0.0.2

network= 20.0.0.0

network= 30.0.0.0

net= 192.168.1.0
0/0= 192.168.1.1

net= 192.168.2.0
0/0= 192.168.2.1

PC-PT PC0
ip= 192.168.1.2
DG= 192.168.1.1

PC-PT PC1
ip= 192.168.1.3
DG= 192.168.1.1

PC-PT PC2
ip= 192.168.2.2
DG= 192.168.2.1

PC-PT PC3
ip= 192.168.2.3
DG= 192.168.2.1

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	PC0	PC2	ICMP		0.000	N	0	(ec
	Successful	PC1	PC3	ICMP		0.000	N	1	(ec
	Successful	PC3	PC0	ICMP		0.000	N	2	(ec
	Successful	PC2	PC1	ICMP		0.000	N	3	(ec

```

Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, Serial3/0
C    20.0.0.0/8 is directly connected, Serial2/0
O    30.0.0.0/8 [110/128] via 10.0.0.2, 00:34:57, Serial3/0
      [110/128] via 20.0.0.2, 00:34:57, Serial2/0
C    192.168.1.0/24 is directly connected, FastEthernet0/0
O    192.168.2.0/24 [110/65] via 10.0.0.2, 00:39:27, Serial3/0

```


Discussion: OSPF dynamically updates routing tables, allowing routers to select the shortest path between networks. Unlike static routing, OSPF automatically adapts to network changes, improving efficiency and fault tolerance. The experiment demonstrated neighbor relationships, route propagation, and successful inter-network communication. Observing routing tables and testing connectivity confirmed that OSPF correctly distributes network information across routers.

Experiment No: 04

Experiment Name: Implementation of a Network using EIGRP protocol.

Introduction: This experiment demonstrates the implementation of the Enhanced Interior Gateway Routing Protocol (EIGRP) in a network designed in Cisco Packet Tracer. EIGRP is a dynamic routing protocol that allows routers to share routing information efficiently and determine the best path for data delivery. The objective is to configure EIGRP, verify neighbor relationships, and ensure seamless communication between multiple network segments.

Software Requirement:

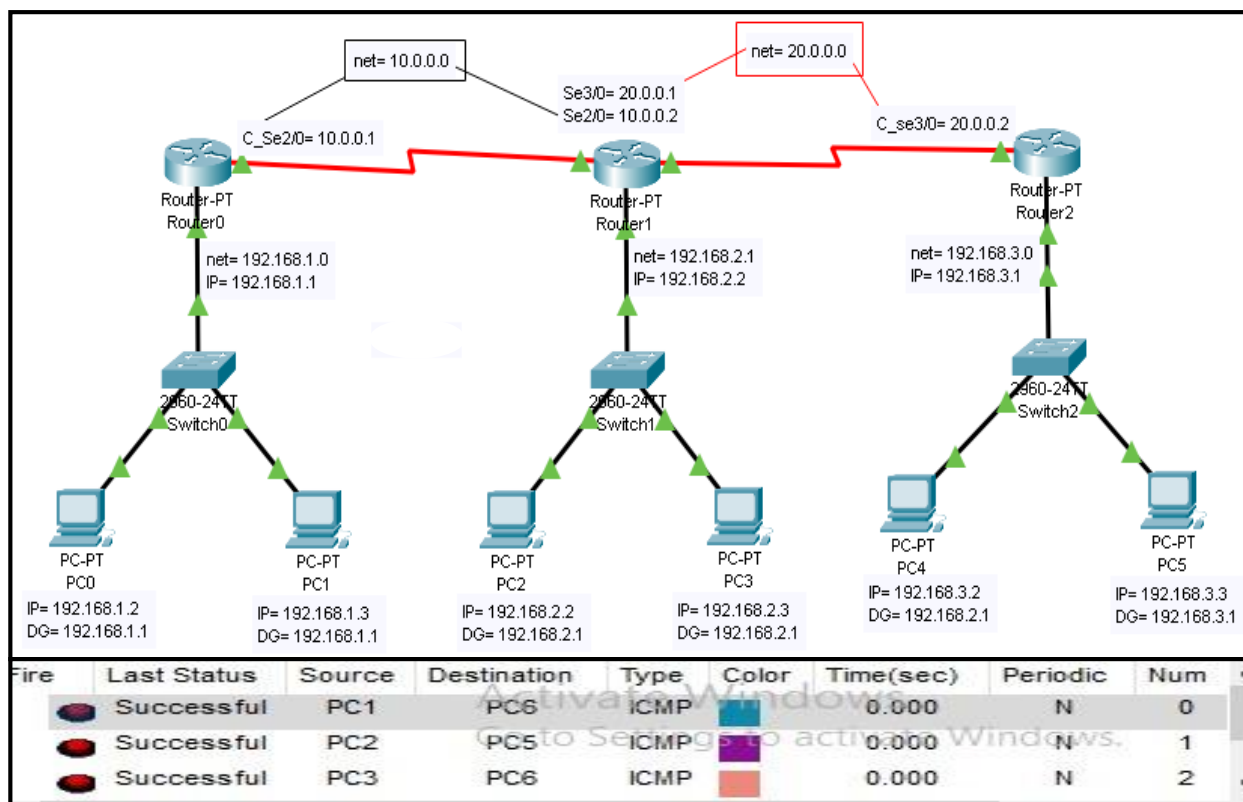
Cisco Packet Tracer – for designing, configuring, and testing the hybrid topology.

Procedure:

- Launch Cisco Packet Tracer and create a new project.
- Place routers, switches, and PCs in the workspace.
- Connect devices using appropriate cables.
- Assign IP addresses to all router interfaces and PCs.
- Access router CLI and enter privileged mode:

Router0	Router1	Router2
Router>enable Router#configure terminal Router(config)#router eigrp 1 Router(config-router)#network 192.168.1.0 0.0.0.255 Router(config-router)#network 10.0.0.0 0.255.255.255	Router>enable Router#configure terminal Router(config)#router eigrp 1 Router(config-router)#network 192.168.1.0 0.0.0.255 Router(config-router)#network 10.0.0.0 0.255.255.255	Router>enable Router#configure terminal Router(config)#router eigrp 1 Router(config-router)#network 192.168.1.0 0.0.0.255 Router(config-router)#network 10.0.0.0 0.255.255.255

- Test basic connectivity using the ping command.
- Send a message using the Simple PDU tool to confirm end-to-end communication.

Output:

Discussion: EIGRP automatically shares routing information between routers, enabling efficient path selection and network scalability. Unlike static routing, EIGRP dynamically adapts to topology changes, reducing downtime. Observing neighbor relationships and routing tables confirmed that EIGRP correctly propagated routes, and successful ping tests demonstrated end-to-end connectivity across the network.